

EXPERIMENT-3

8-BIT MULTIPLICATION

AIM: To write an assembly language program to implement 8-bit multiplication using 8085 processor.

ALGORITHM:

1. Start the program by loading a register pair with the address of memory location.
2. Move the data to a register.
3. Get the second data and load it into accumulator.
4. Add the two register contents.
5. Increment the value of the carry.
6. Check whether the repeated addition is over.
7. Store the value of product and the carry in the memory location.
8. Halt.

PROGRAM:

MNEMONICS	EXPLANATION
LDA 2200	load the accumulator with the first number in the address 8500.
MOV E,A	Move the data from accumulator to 'E' register.
MVI D,00	Move the immediate value 00 into.
LDA 2201	load the accumulator number in the address 2201.
MOV C,A	move the data from accumulator to 'C' register.
LXI H,0000	load the immediate value 0000 into the HL register pair.
BACK: DAD D	Back: label for the loop DADD: Add the value in register D.
DCR C	Decrement register E by 1
JNZ BACK	In register E, is not 0, jump back to the beginning of the loop.
SHLD 2202	Store the value of the HL register pair at memory address 2202
Halt	Halt.

Input:

ADDRESS	DATA
2200	4
2201	2

Output:

ADDRESS	DATA
2202	8

Result: Thus the program was executed successfully using 8085 processor simulator.

FileResetAssemblerDebugHelp

Registers

A	03
BC	00 00
DE	00 05
HL	00 0F
PSW	00 00
PC	42 16
SP	FF FF
Int-Reg	00

Flag

S	0
Z	1
AC	0
P	1
C	0

Decimal - Hex Conversion

Decimal	Hex
0	0
To Hex	To Dec

I/O Ports

0	-	+	00
Update Port Value			

Memory

2201	-	+	03
Update Memory			

Load me at

1 LDA 2200

2 MOV E, A

3 MVI D, 00

4 LDA 2201

5 MOV C, A

6 LXI H, 0000

7 BACK: DAD D

8 DCR C

9 JNZ BACK

10 SHLD 2202

11 HLT

12

GNUSim8085 - 8085 Microprocessor Simulator

DataStackKeyPadMemoryI/O Ports

Start2202OK

Address (Hex)	Address	Data
089A	2202	15
089B	2203	0
089C	2204	0
089D	2205	0
089E	2206	0
089F	2207	0
08A0	2208	0
08A1	2209	0
08A2	2210	0
08A3	2211	0
08A4	2212	0
08A5	2213	0
08A6	2214	0
08A7	2215	0
08A8	2216	0
08A9	2217	0

Line NoAssembler Message

0Program assembled successfully